

MIKROELEKTRONIKA

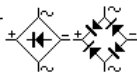
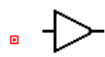
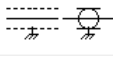
The following table covers almost every circuit symbol you will need. This is the English/American version of each symbol. The European version of some symbols is slightly different and are shown further down the page.

a.	b.	c.	d.	e.	f.	g.	h.	i.	j.	k.
fuse	lamp	neon lamp	VDR resistor	cell	quartz crystal	ammeter or voltmeter	AC voltage	switch	double pole switch	4 position switch

10.1.

CIRCUIT SIMBOLS by Talking Electronic

AC current:		voltage:	
ALTERNISTOR TRIAC		Main Terminal1	
A TRIAC and 33 - 43v DIAC		Gate	
Main Terminal 2			
Ammeter (amp meter)			
AND Gate			
AND Gate			
Antenna balanced			
Antenna Loop, Shielded			
Antenna Loop, Unshielded			
Antenna unbalanced			
Attenuator, fixed (see Resistor)			
Attenuator, variable (see Resistor)			

Battery	
Bilateral Switch (DIAC)	
Bridge Rectifier (Diode Bridge)	
BUFFER (Amplifier Gate)	
BUFFER (Amplifier Gate)	
Buzzer	
Capacitor feedthrough	
Capacitor non-polarised	
Capacitor polarised (see electrolytic)	
Capacitor Variable	
Cathode Ray Oscilloscope – CRO	
Cavity Resonator	
Cell	
Circuit Breaker	
Coaxial Cable	
Connectors	      
CRO – Cathode Ray Oscilloscope	
Crystal Microphone (Piezoelectric)	
Crystal Piezoelectric	
Darlington Transistor	
DC current:	
voltage:	
Delay Line	

DIAC
(Bilateral Switch)



Diode



Diode Bridge
(Bridge Rectifier)



Diode - Gunn



Diode - Light Emitting
(LED)



Diode
Photo Sensitive



Diode
Photovoltaic



Diode - Pin



Diode - Varactor



Diode - Zener



Earpiece
(earphone,
crystal earpiece)



Electroluminescence



Electret Microphone
(Condenser mic)



Electrolytic
(Polarised Capacitor)

alternate symbols:
(positive on top)



Electrolytic - Tanatalum
positive end
black band or
chamfer



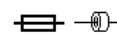
Exclusive-OR Gate
(XOR Gate)



Exclusive-OR Gate
(XOR Gate)



Ferrite Bead



Field Effect
Transistor
(FET) n-channel
also: N-Channel J FET

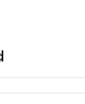
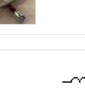


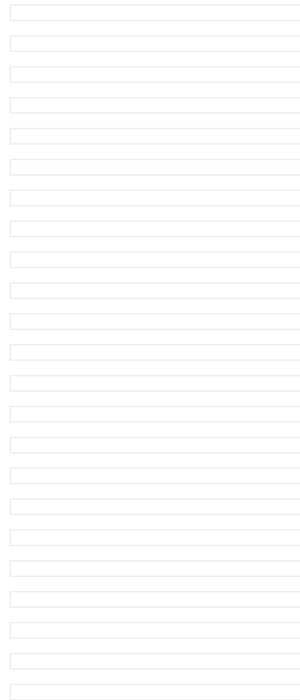
Field Effect
Transistor
(FET) p-channel
also: P-Channel J FET



Flashing LED
(Light Emitting Diode)
(Indicates chip inside LED)



Fuse	
Galvanometer	
Globe	
Ground Chassis	
Ground Earth	
Headphone	
Heater	
IC Integrated Circuit	
Inductor Air Core	
Inductor Iron Core or ferrite core	
Inductor Tapped	
Inductor Variable	
Integrated Circuit	
Inverter (NOT Gate)	
Jack Co-axial	



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NOTES

Here are a few notes on the symbols above.

Fuses (10.1a) have single role in a circuit – to detect excess current and protect the device. In most cases the excess current flows when a higher voltage is present but a fuse cannot detect the voltage – it can only detect when a higher current flows. The higher voltage causes the higher current to flow and this triggers the action of “blowing the fuse.” Of course, when a component fails, a higher current can flow and this will also “blow the fuse.”

Fuses come in all sizes and ratings (current flow) and it is important to know that the size of the wire inside a fuse does not necessarily indicate the current rating.

The wire inside can be made from copper and plated to protect it from oxidizing or it can be a low temperature material that needs to be a larger diameter.

The wire can also be wound in a spiral and formed into a spring. The end of the spring sits in a dab of solder and when the spring heats up, the solder melts and the spring separates from the other end.

This is called a DELAY FUSE.

Other forms of delay fuse consist of a wire joined at the centre by a dab of solder and others are made of low-temperature-melting material.

Some pieces of equipment use expensive fuses and whenever a fuse is damaged, you must decide if the problem is a major or minor fault.

Sometimes a fuse can go open-circuit for no apparent reason. It can “wear-out.”

For instance, some equipment takes a very high current when it is turned on and you will see the fuse heat up and stretch and dip in the middle. This causes strain on the fuse and eventually the wire oxidizes to a point where it finally “burns out.”

The equipment is not faulty and it is just a matter of replacing the fuse.

Sometimes the fuse completely explodes and the glass is thrown all over the chassis. This indicates a short-circuit in the power supply and most often one or more of the diodes must be replaced.

The fuse can also go off with a “bang” and the inside of the glass is coated with “silver.” This also indicates a diode is damaged in the power supply. Generally 2 or 4 diodes are damaged.

If the fuse is damaged beyond recognition, you will not know if it is a delay fuse or a normal fuse.

The current-rating on the end-cap can sometimes help you.

For instance, if a fuse is rated at 4A, you will need to replace it with a 4 amp normal fuse or 3.15 amp slow-blow.

When fuses are rating at 100mA to 250mA, they are very delicate and will not accept the slightest overload. When replacing this type of fuse, it is necessary to determine if the equipment is drawing a heavy current when turning on or if a fault exists in the power supply. Sometimes the switch can cause the problem if it is not making contact fast enough.

Replace the fuse and watch it as someone else turns on the equipment. If the fuse burns out immediately, a short exists. If the fuse glows red and burns out, the equipment is drawing too much current during turn-on. This may be due to devices you have added to the equipment or operation on a slightly higher voltage. You can try a fuse with a slightly higher rating to see if the fault is fixed.

Never replace a 100mA fuse with a 1 amp fuse. The 1 amp fuse will never “blow” and if the transformer is being overloaded, the transformer will simply “cook.”

Lamps (10.1b) Ordinary electric light globes heat a coil of tungsten wire inside a glass bulb that has an inert gas such as argon. The resistance of the filament depends on the temperature it is heated to. It can be ten to twenty times higher than when it is cold.

A neon lamp (10.1c) contains a gas (such as neon) and this gas gives off a glow when a high voltage is applied to two plates. This glow occurs at about 70v to 90v and a resistor must be used in series to prevent the voltage rising higher than required by the lamp. To put this more accurately, the resistance of the neon lamp reduces when it “strikes” and a high current will flow. To limit this current a “current limit” resistor is needed.

VDR (10.1d) The resistance of a VDR depends on the voltage across it. A VDR is also called a VARISTOR. Its resistance is high until a critical value of voltage and the resistance suddenly drops. They are used as voltage protection devices. If they, for example, see a voltage higher than 220V, their resistance decreases and this “soaks” the excess voltage. Their response time is only a few 10's of nanoseconds.

The symbol for a single DC cell is shown in 10.1e.

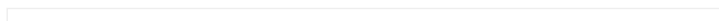
A Quartz crystal is shown in 10.1f. It is a thin sheet of quartz material between two metal plates and packaged in a metal case. Quartz crystals are commonly used as the reference for an oscillator circuit, such as a clock source in microprocessor designs.

An instrument for measuring current (A) and voltage (V) is shown in 10.1g. This symbol dates from the time when analog instruments with a needle were used. The symbol remains the same, although digital instruments have replaced analog devices.

AC voltage symbol is shown in 10.1h. The shape of the wave is shown in the symbol. It can be sine-wave or saw-tooth or square-wave.

The simplest form of switch device is displayed in 10.1i. Because of the wide range of switches, there are many different types in use. For example, a two pole switch (10.1j) has two operating positions, in one position it connects points 1 and 2, and in the other it connects points 1 and 3.

There are switches with more operating positions. 10.1k is an example of a rotary switch with four positions.



Momentary switches, or push buttons have a built-in spring, which makes the switch conduct only while it is being pressed (your standard doorbell has this kind of switch).

Four diodes in a single case is called a BRIDGE. Two pins are marked with sine waves, used to connect to the AC voltage and two marked with “+” and “-“

RELAY When an electromagnet receives sufficient voltage on points 4 and 5, connection between points 2 and 3 is opened, and at the same time points 3 and 1 are closed. A relay is actually an electromagnetic switch.

Symbols for a receiving and transmitting antenna are shown.

Grounding symbols Grounding and common ground aren't the same thing, but if both exist in a circuit, they are always connected to each other. With electronic devices housed in a metal case, grounding is connected to the metal housing.

Schematic symbols representing logic gates and different digital integrated circuits are shown above. It should be kept in mind that basic logic gates (AND, OR, XOR, Inverter, etc.) aren't manufactured as single standalone components. They are always integrated in groups in an IC, but for the sake of clarity, they are represented as separate blocks. These components require a DC voltage, which may or may not be represented on the schematic. These voltages might be different depending on the internal structure and technology used between different family types. Detailed info on this can be found in the component's datasheet provided by the manufacturer.



10. Introduction to Other components by MikroElektronika is licensed under a Creative Commons Attribution 4.0 International License, except where otherwise noted.